

Job Proposal

Waze User Churn Prediction Case Study

I built this project as an end-to-end machine learning case study focused on predicting monthly user churn from behavioral data.

What I handled

- Data quality checks and preparation
- Exploratory analysis of user activity and churn
- Statistical testing using Welch's t-test
- Behavioral feature engineering
- Train/validation/test workflow
- Logistic Regression baseline
- Random Forest and XGBoost tuning
- Validation-based model selection
- Threshold analysis for recall/precision trade-offs
- Final held-out test evaluation
- Feature importance and model persistence

Result

XGBoost was selected as the ensemble model based on validation recall. On the untouched test set, the default 0.50 threshold achieved 0.8168 accuracy, 0.4555 precision, 0.1716 recall, 0.2493 F1, and 0.7318 ROC AUC. A validation-selected threshold of 0.223 increased test recall to 0.5247 and F1 to 0.4266, with precision of 0.3595.

Client Deliverables

The project package includes the complete Jupyter notebook, a concise executive summary, selected visualizations, and this proposal. The analysis is structured so that the workflow, assumptions, validation strategy, and model limitations are visible rather than hidden.

Important Modeling Note

The dataset is educational and the scenario is fictitious. The model results should not be presented as real Waze performance. For a production project, I would validate prediction-time feature availability, test on fresh data, and define the operating threshold from the actual business cost of false positives and missed churners.

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